Tamer Mahmoud

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Senior Software Engineer | Full-Stack Developer | Cloud & Automation Engineer

Software Engineer with 10+ years of experience working across backend, frontend, and cloud systems. Strong background in building and operating production APIs, event-driven services, and data pipelines in AWS and GCP. Experience spans the full production lifecycle, from local development and code review through deployment, environment and infrastructure management, monitoring, debugging, and ongoing feature iteration in remote teams.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, PHP, Go
Frontend Frameworks & Libraries: React, Next.js, Tailwind CSS, Ember.js, Radix UI, Angular
Backend Development: Node.js, Prisma, REST APIs, ElasticSearch (ES)
Database Technologies: PostgreSQL (Postgres), MySQL, MSSQL, MongoDB (with Mongoose), Sequelize ORM
Cloud & Infrastructure: AWS (Lambda, SQS, EventBridge), GCP, Docker, Serverless Framework
Monitoring & CI/CD: CloudWatch, Datadog, CircleCI, GCP Logging
Design & Collaboration Tools: Figma, LaunchDarkly, Courier
AI & GenAI Tools: OpenAI API, Embeddings, Prompt Engineering, Large Language Models (LLMs)
Testing Frameworks & Libraries: QUnit, Cypress, React Testing Library (RTL), Jest, Mocha, Chai

PROFESSIONAL EXPERIENCE

Software Engineer | Expel — 12/2022 – 06/2025

- Built and supported backend and frontend features for alert, investigation, and incident workflows, updating Node.js APIs, writing and tuning PostgreSQL queries, shaping API responses, and consuming those APIs in React and Ember.js to deliver real-time security views for internal teams and customers.
- Worked on Go-based backend API services and Python-based automation and worker jobs triggered through Pub/Sub, handling event-driven processing and operational tasks outside of request and response flows.
- Deployed and operated services using the Serverless Framework, defining infrastructure in YAML including Lambda functions, Firehose delivery streams, Athena queries, Glue jobs, S3 buckets, and IAM roles and policies. Investigated and resolved permission and configuration issues when deployments or runtime behavior did not work as expected.
- Set up and maintained CloudWatch logging, metrics, and alarms for backend services. Used CloudWatch Logs Insights and the AWS CLI to inspect logs, review metrics, and validate alarm behavior. Configured alerting flows where CloudWatch alarms triggered SNS and delivered notifications into Slack, and regularly adjusted alarms to reduce noise.
- Queried data stored in S3 using Athena to investigate alert and incident data, validate pipeline output, and support reporting by writing and refining SQL queries over large datasets in AWS.
- Worked with data in GCP using Cloud SQL and BigQuery to support application features and analysis needs, writing SQL queries and validating results used by backend services and internal tooling.

Full Stack Developer | Sales Boomerang — 04/2022 – 10/2022

- Worked across frontend and backend product features, building user interfaces in Angular and React with TypeScript, implementing Node.js APIs with Sequelize ORM for data access, and coordinating API behavior and response shapes so frontend features could load, update, and display data correctly.
- Updated REST APIs and database queries to support batched request and response patterns, allowing the frontend to request and receive multiple records in a single call while keeping validation and data shaping consistent across features.
- Deployed and supported backend infrastructure using CloudFormation defined in YAML, working with AWS services including Lambda, SNS, and SQS. Used CloudWatch logs and alarms to verify releases and troubleshoot issues caused by configuration or infrastructure changes.

Web Consultant | Rural Sourcing, Inc. 08/2021 – 04/2022

- Worked on breaking apart a legacy PHP application by moving specific functionality into separate services, updating PHP endpoints and React components so features could be developed and deployed independently instead of through the full monolith.
- Added and updated API endpoints used by the React frontend, wiring requests to backend PHP logic and database queries so data could be fetched, updated, and displayed correctly.
- Wrote Python scripts to pull aggregated data from existing data sources, run queries for reporting use cases, and output results for internal teams instead of manually running queries.
- Used Docker to run the application, services, and scripts locally, matching production dependencies so bugs could be reproduced and tested without relying on shared environments.
- Investigated and fixed API issues by tracing failing requests, identifying where responses were incorrect or missing, and correcting backend logic so endpoints returned expected data instead of errors or empty results.

Software Engineer | Gallaudet University 10/2012 – 08/2020

- Migrated large datasets from Microsoft Access to Microsoft SQL Server by designing new schemas, writing data migration scripts, validating records, and resolving data inconsistencies so existing applications could continue operating without interruption.
- Built REST APIs that connected frontend pages to backend logic and databases, handling create, read, update, and delete operations so multiple systems could share and update data reliably.
- Implemented a real-time video messaging feature using WebRTC and RecordRTC, handling video capture, recording, and playback to support direct video communication inside the application.
- Built internal tools such as online assessment and data-entry forms that automated repetitive tasks, replacing manual spreadsheets and email-based workflows with web-based systems tied directly to application databases.
- Modernized older web applications by introducing React for frontend views and Node.js for backend services, incrementally replacing legacy patterns while keeping existing functionality working.
- Applied accessibility standards across custom CMS features and web interfaces, including keyboard navigation, screen reader support, semantic markup, and color contrast, to meet WCAG 2.1 AA requirements.

PROJECTS

Software Engineer Portfolio

Built a responsive portfolio using React, Vite, and Tailwind CSS, featuring dark/light mode, animated transitions, and a starfield background. Integrated a custom AI chatbot powered by OpenAI and Python, using embeddings over resume and project content to answer common recruiter and developer questions.

GitHub · Live Demo

EDUCATION

Bachelor of Science in Information Technologies | Rochester Institute of Technology